

2(a)	X (nucleolus) synthesis / AW of, ribosomal subunits / ribosomes ; A synthesis of organelles that are the sites of protein synthesis Y (chromatin) contains gene(s), coding for the polypeptide(s) / AW ; A site, of transcription / for production of mRNA	2
2(b)(i)	primary ;	1
2(b)(ii)	any one from: carboxylic acid / carboxyl, group ; A both have C=O contain carbon / hydrogen / oxygen ;	1

Question	Answer	Marks
2(b)(iii)	any three from: ref. to role in / forms part of, overall 3D shape ; A tertiary / quaternary structure A forms part of hydrophobic region of ghrelin allows ghrelin to bind to receptor ; allows, ghrelin / part of ghrelin, to have a complementary shape / be complementary (to the receptor) ; A description suggested detail ; e.g. interacts with hydrophobic region of the receptor molecule (for binding) AVP ; e.g. help interaction with / embeds in, hydrophobic portion of the cell surface membrane <i>in context of bringing a complementary part of the molecule closer to receptor for binding</i> suggestion that fatty acid chain allowed, incorporation into / attachment to, a transport molecule in plasma to reach target cell ref. to effect of binding triggering reactions inside the cell ;	3
2(c)(i)	1.8% ; working ; $\frac{30 \times 3}{5000} \times 100$ <i>codons for 29 amino acids (including start met) and a stop codon</i> <i>allow 1 mark if only 28 or 29 instead of 30, so 1.7% answer obtained</i>	2

Question	Answer	Marks
2(c)(ii)	any two from: introns / non-coding regions, removed / AW ; exons / coding regions, are joined together ; gene splicing / RNA splicing ; R DNA splicing AVP ; e.g. guanine (cap) added (to 5' end) poly A tail / many As, added (to 3' end)	2

Question	Answer	Marks
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